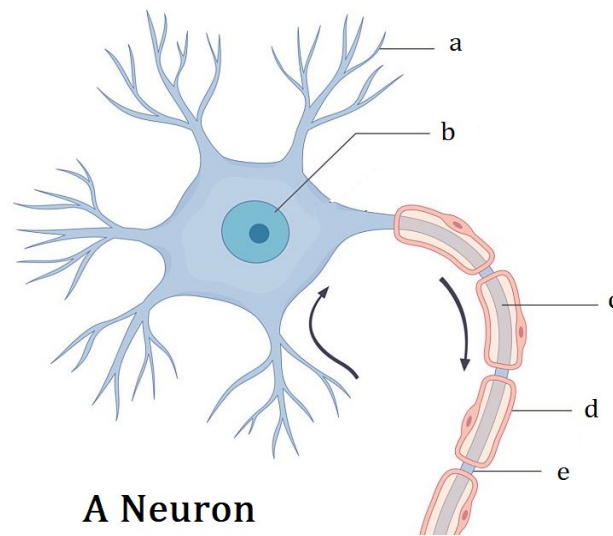


Exemplar #1

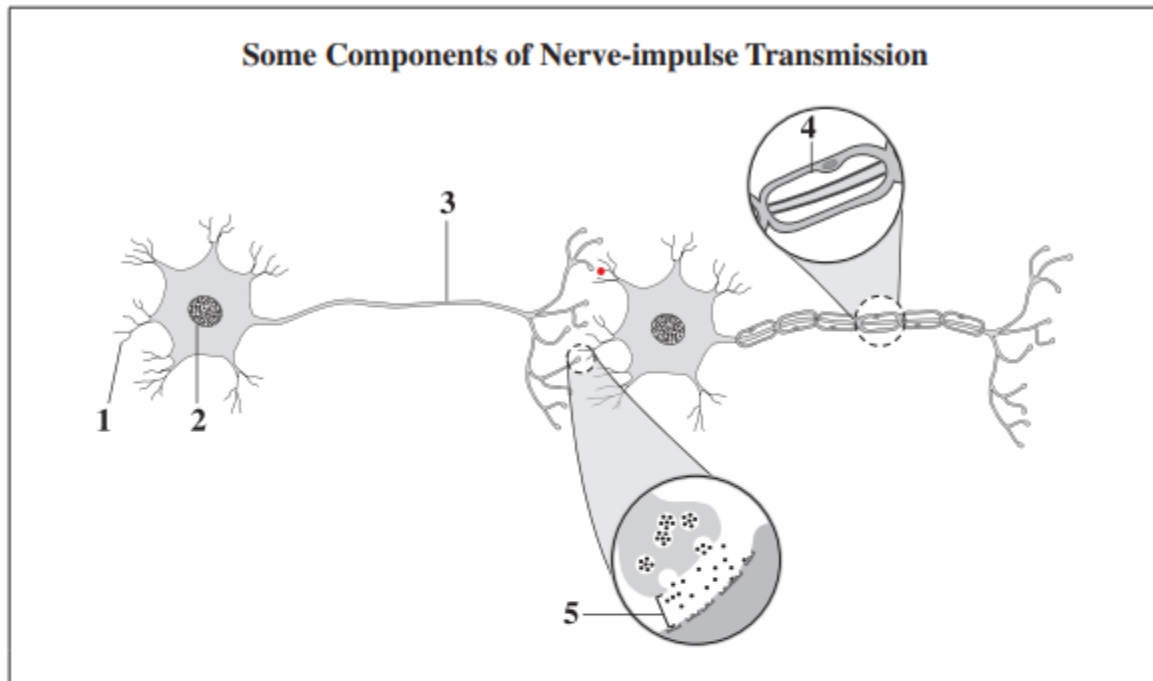
Charcot-Marie-Tooth disease (CMT) is a neurological disorder characterized by nerve damage that causes loss of sensation. The two most common forms of this disorder are CMT1 and CMT2. CMT1 is caused by a genetic mutation that results in abnormal structure and function of the myelin sheath, and CMT2 is caused by a genetic mutation that leads to the abnormal functioning of the axon.



- 1) What type of neuron is most directly impacted by CMT?
- 2) Identify the structure in the diagram above that is affected by CMT1.
- 3) Identify the structure in the diagram above that is affected by CMT2.
- 4) Based on the information provided, suggest one other possible symptom that might result from CMT.

Exemplar #2

Use the following information to answer numerical-response question 1.



Numerical Response

1. Match four of the components of nerve-impulse transmission numbered above with the descriptions given below.

Number:	_____	_____	_____	_____
Description:	Area into which neurotransmitter is released	Increases speed of transmission	Carries nerve impulses away from the cell body	Receives information from other neurons

(Record all **four digits** of your answer in the numerical-response section on the answer sheet.)

Exemplar #3

Lignocaine is an anesthetic used to block pain during dental procedures. Lignocaine diffuses into neurons and works by blocking sodium channels.

- 1) Does lignocaine disrupt active or passive transport within the neuron?
- 2) Explain how blocking sodium channels would prevent an individual from feeling pain in a particular area.
- 3) Predict the membrane potential of a neuron that has been altered by lignocaine. Explain your answer.

Exemplar #4

Use the following information to answer question 2.

Researchers have found flavour-modulating chemicals that increase the likelihood that sugar molecules will bind to taste receptors. Small amounts of sugar taste slightly sweet. However, the combination of the same amount of sugar and the flavour modulators intensifies the perception of sweetness.

2. Which of the following rows identifies two processes in neurons that could account for the intensifying of sweetness when sugar is combined with flavour modulators?

Row	Process 1	Process 2
A.	A larger number of neurons is stimulated.	Impulses occur with greater frequency.
B.	A larger number of neurons is stimulated.	The response threshold of the neurons is increased.
C.	Stimulated neurons produce a stronger impulse.	Impulses occur with greater frequency.
D.	Stimulated neurons produce a stronger impulse.	The response threshold of the neurons is increased.

Exemplar #5

Adrenoleukodystrophy (ALD) is a genetic disorder that causes damage to the myelin sheath of neurons in the brain. Some symptoms of ALD include:

1. Decreased vision
2. Decreased hearing
3. Decreased learning ability
4. Decreased balance/coordination

- 1) Match each of the symptoms of ALD listed above with the region of the brain associated with it given below:

Cerebellum

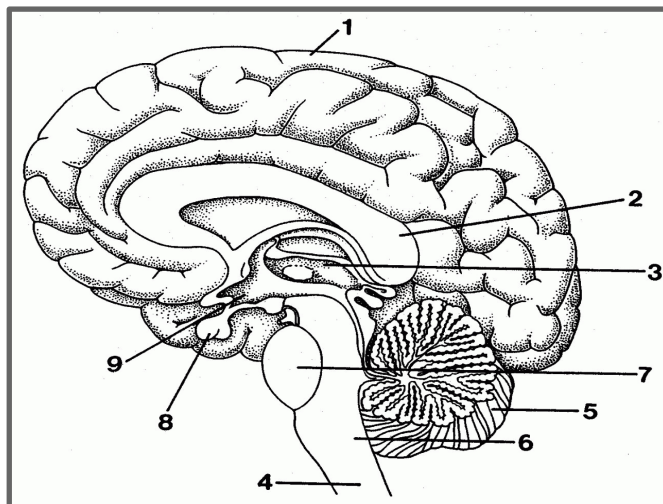
Frontal Lobe

Occipital Lobe

Temporal Lobe

- 2) Does ALD affect the white or grey matter of the brain? Explain how you know.

- 3) Identify the cerebellum in the following diagram of a human brain:



Exemplar #6

Exposure to certain foods or environmental agents can cause a severe allergic reaction known as anaphylaxis. Common symptoms of anaphylaxis include a decrease in blood pressure and constriction of bronchioles in the lungs.

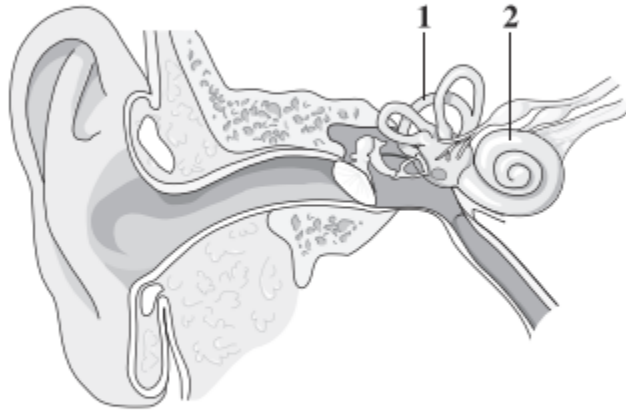
- 1) Which division of the autonomic nervous system is responsible for producing effects similar to those of anaphylaxis?
- 2) Identify the region of the brain that is responsible for regulating aspects of the autonomic nervous system such as circulation and breathing.

Exemplar #7

Use the following information to answer question 6.

Pendred syndrome is a genetic disorder that can result from the development of an abnormal cochlea.

The Human Ear



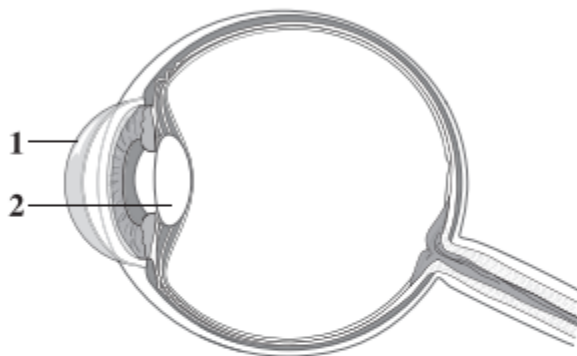
6. The structure of the human ear that develops abnormally in a person with Pendred syndrome is numbered
- A. 1, and it is responsible for hearing
 - B. 1, and it is responsible for balance
 - C. 2, and it is responsible for hearing
 - D. 2, and it is responsible for balance

Exemplar #8

Use the following information to answer question 5.

Laser eye surgery involves the use of a laser to remove small amounts of eye tissue. In one type of laser eye surgery, vision is improved by reshaping the cornea.

The Human Eye



5. Which of the following rows identifies the structure that is reshaped during the type of laser eye surgery described above and **best** describes the function of the structure identified?

Row	Structure	Function
A.	1	Focuses images on the retina
B.	1	Bends light toward the pupil
C.	2	Focuses images on the retina
D.	2	Bends light toward the pupil